

Niranjana (Falgu) River Rejuvenation Project

Mission- To make Falgu River rejuvenated

Niranjana (Falgu) River

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New feature

Present

Catchment Area

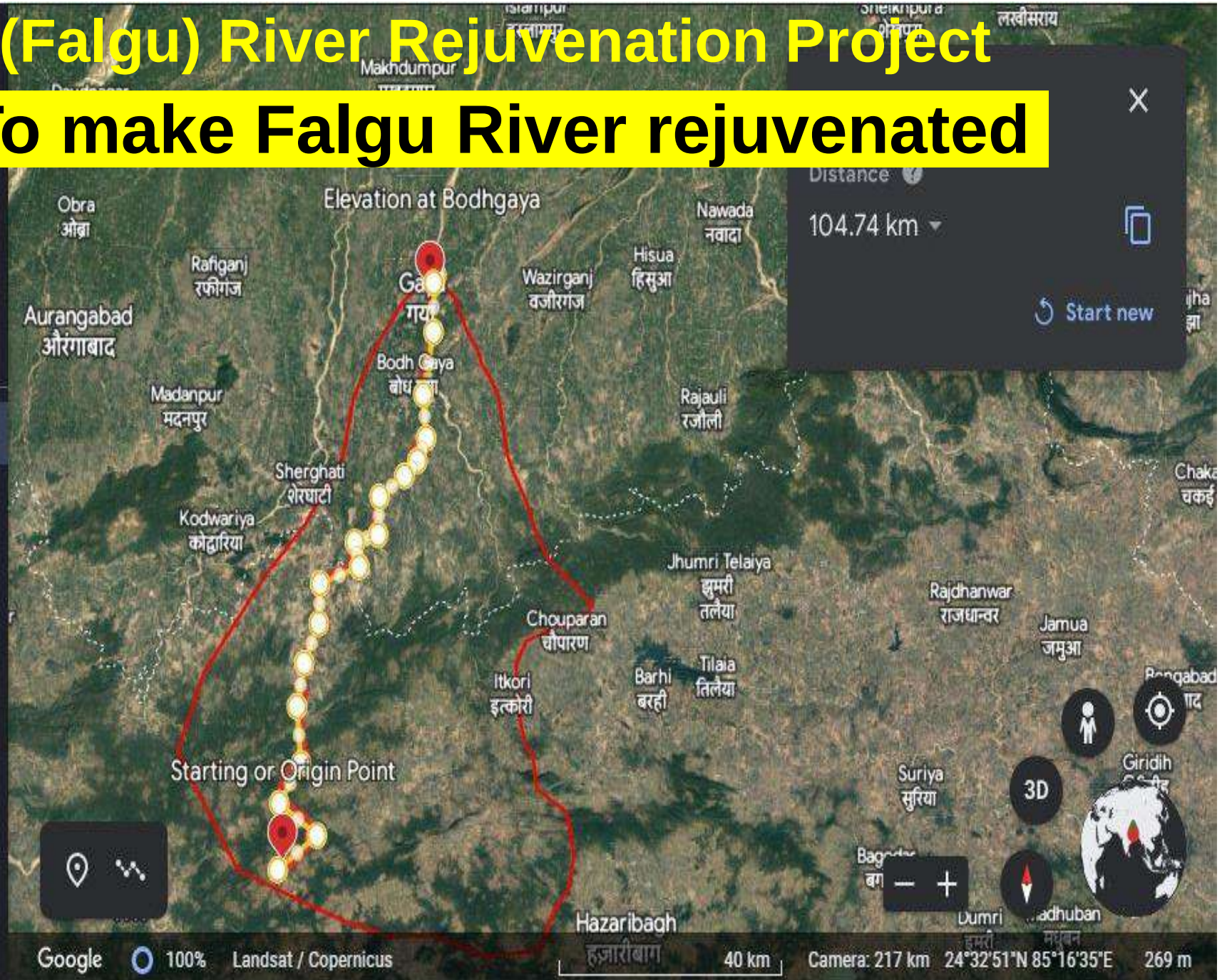
Elevation at Bodhgaya

Falgu River Stretch 1

Falgu River Stretch 2

Falgu River Stretch 3

Starting or Origin Point



Niranjana (Falgu) River

- River Niranjana (Falgu), has religious and cultural significance for both Buddhists and Hindus.
- River has dried and is dying. As per literature, this river was perennial.
- Originates from Belgadha and Lavalong of Simaria block under Chatra district of Jharkhand. Flowing silently in a narrow channel between the rocky hills, this river becomes bigger and wider when it enters Bihar through Hunterganj. This river is known as Falgu in Bihar

Letter sent by	:	Niranjana (Phalgu) River Recharge/ Rejuvenation Mission
Concerned Organization	:	Niranjana (Phalgu) River Recharge Mission
	:	Bodhgaya, Gaya, Bihar, India - 824231
Origin	:	Belgadha and Lavalong of Simaria block under Chatra district of Jharkhand
Length of River	:	260 km
Length covered in proposal	:	130 km
Site Visit Done on 15 th to 17 th March, 2023		

Characteristics of River are as under:

Antah Sataha River

- The nature of Falgu river is Antah Sataha- means flowing under the riverine sands.
- Medium to fine grained well sorted river bed
- Ground water fed river during lean season
- Flash flood during monsoon season
- Ground water in the riverbed as told 10-15 m bgl at present

Catchment Area

- Catchment area= tentatively 3000 sq km (Based on Google Earth)
- However large scale topographic mapping using Aerial LIDAR is proposed for catchment area.
- Elevation of Starting Point = 496.20 m
- Elevation at Bodhgaya = 136.44 m
- Average slope 0.00343 = Say 1 in 291

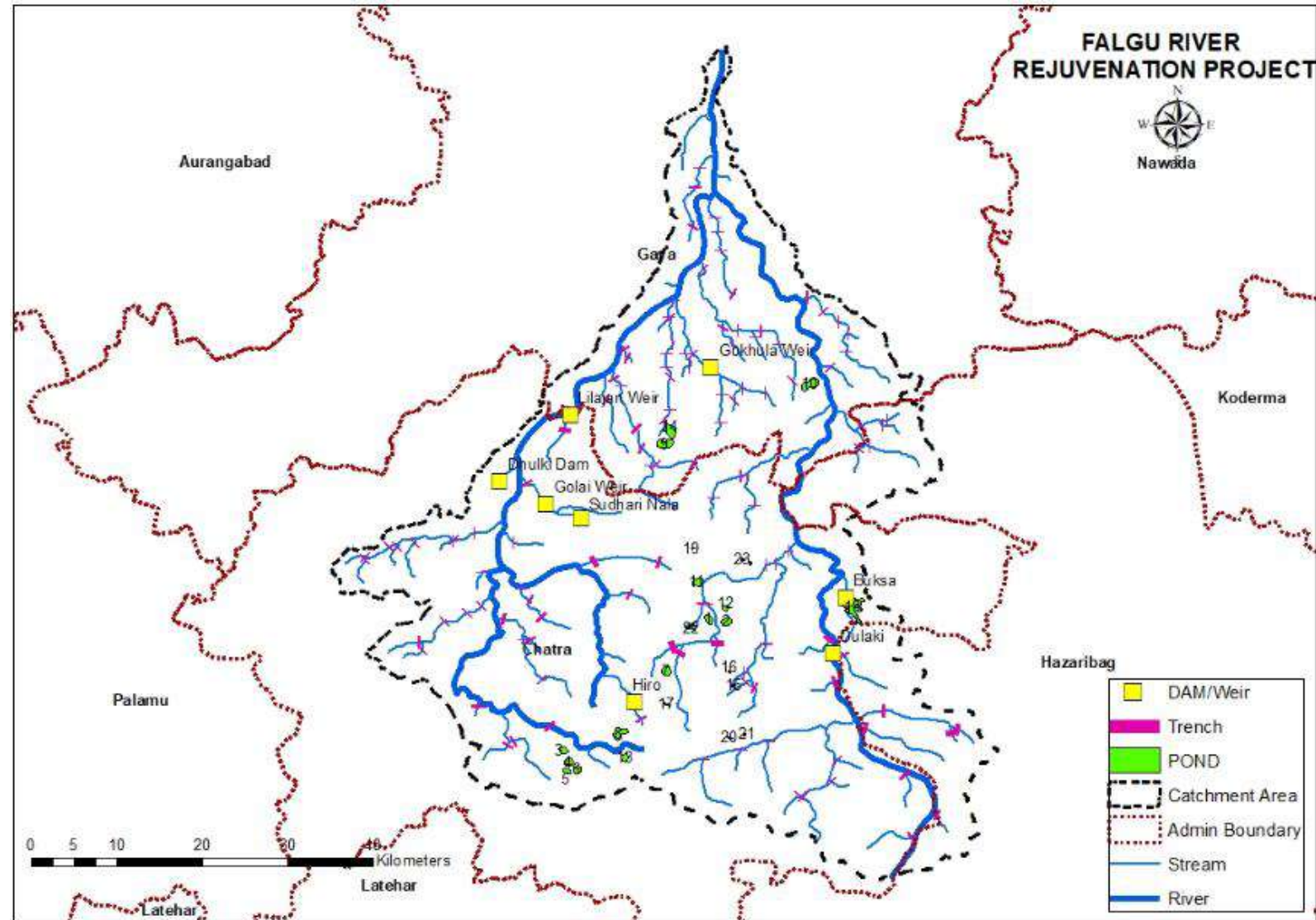


Immediate Measures- For Artificial Ground Water Recharge through Ponds & Trenches at U/S Catchment Area

Tentative Locations of Trenches & Ponds

Pond Area for De-silting

Pond No.	Shape *	Area (m ²)
1	Polygon	712518
2	Polygon	799716
3	Polygon	560644
4	Polygon	621292
5	Polygon	480118
6	Polygon	544854
7	Polygon	842054
8	Polygon	1300486
9	Polygon	1478309
10	Polygon	1603592
11	Polygon	795912
12	Polygon	419943
13	Polygon	507884
14	Polygon	1335904
15	Polygon	81142
16	Polygon	32918
17	Polygon	112208
18	Polygon	3111158
19	Polygon	15221
20	Polygon	65760
21	Polygon	28484
22	Polygon	349765
23	Polygon	92902
24	Polygon	63346
Total Area		15956128



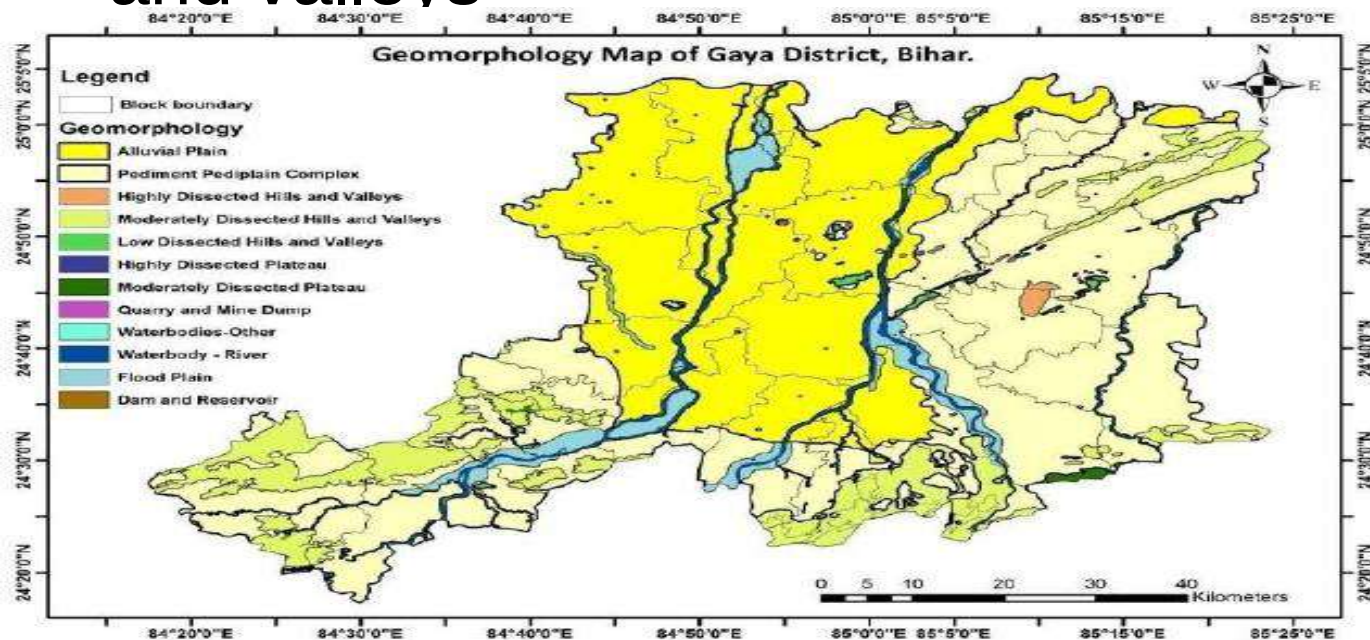
Hydrological Calculations based on District Gaya Daily Rainfall Data (2014 to 2020)

Name of Project :		Niranjana (Phalgu) River Recharge/ Rejuvenation Mission				
Submitted By :		Niranjana (Phalgu) River Recharge Mission				
		Bodhgaya, Gaya, Bihar, India - 824231				
Origin :		Belgadha and Lavalong of Simaria block under Chatra district of jharkhand				
Length of River :		260 km				
Length covered in proposal :		130 km				
Location of First Check Dam :		Kolhua village (Lado Panchayat)				
S. No.	Year	Max. Daily Rainfall (mm)		Yearly Total Rainfall (mm)		Date
1	2014	55.3		877.03		05-Jul-14
2	2015	75.5		738.33		28-Jun-15
3	2016	95.69		1185.72		06-Sep-16
4	2017	64.8		658.2		05-Jul-17
5	2018	35.6		573.1		26-Jul-18
6	2019	104.6		1045.9		29-Sep-19
7	2020	58		1014.1		22-Sep-20
S. No.	Return Period in Yrs.	Design Daily Rainfall (mm)				
		Gumble	Log Pearson	Log Pearson III	Average	
1	25	144	118	133.00	131.67	
2	50	163	127	153.60	147.87	
3	100	182	136	176.00	164.67	

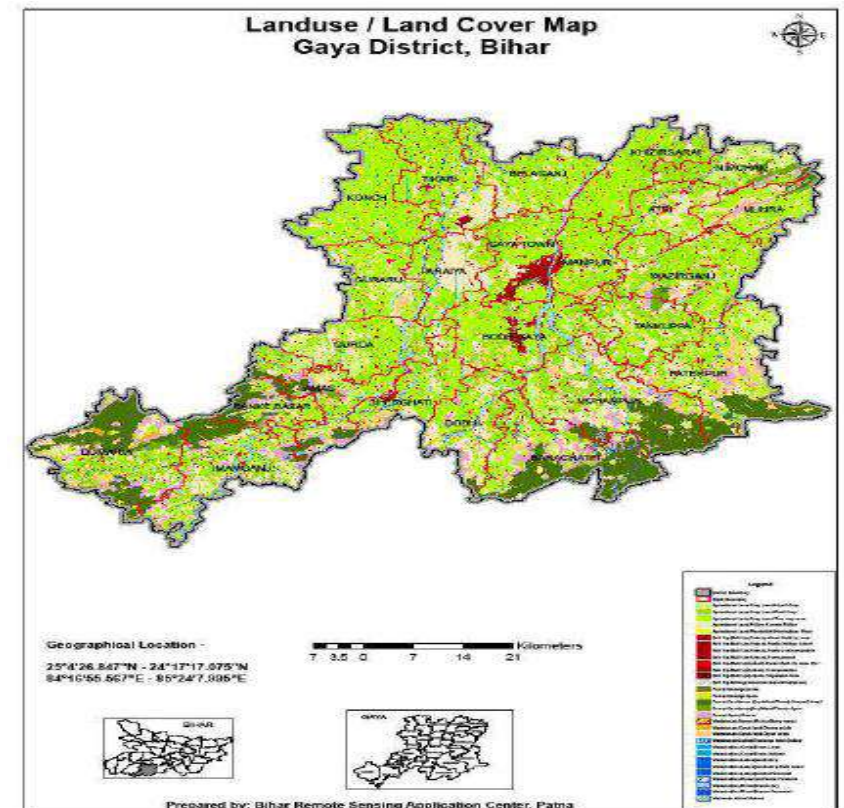
Catchment Area =		3046	Km ²		
		=3046000000m ²			
Rainfall Intensity =		164.67	mm in 24 hours (For 100 Yrs return period)		
		= 0.00190586mm/ sec			
Discharge Q =		5805.26Cumec (m³ /sec)			
Rainfall Intensity =		220.20	mm in 24 hours (Taken Lilajan Irrigation Scheme)		
		= 0.00254861mm/ sec			
Discharge Q =		7763.07Cumec (m ³ /sec)			
Elevation of Starting Point =		496.20 m			
Elevation at Bodhgaya =		136.44 m			
Difference =		359.76 m			
Length of Falgu River Stretch =		104.74 Km			
Length of Falgu River Stretch =		104740.00 m			
Average Slope of the River =		$\frac{359.76 \text{ m}}{104740 \text{ m}}$			
		$=0.00343 = \frac{343}{100000}$			
		$= \frac{1}{291}$ Say 1 in 291			

Geomorphology

- The catchment has alluvial plain and pediplain offers as good ground water storage and reservoirs
- The origin area is dissected hills and valleys

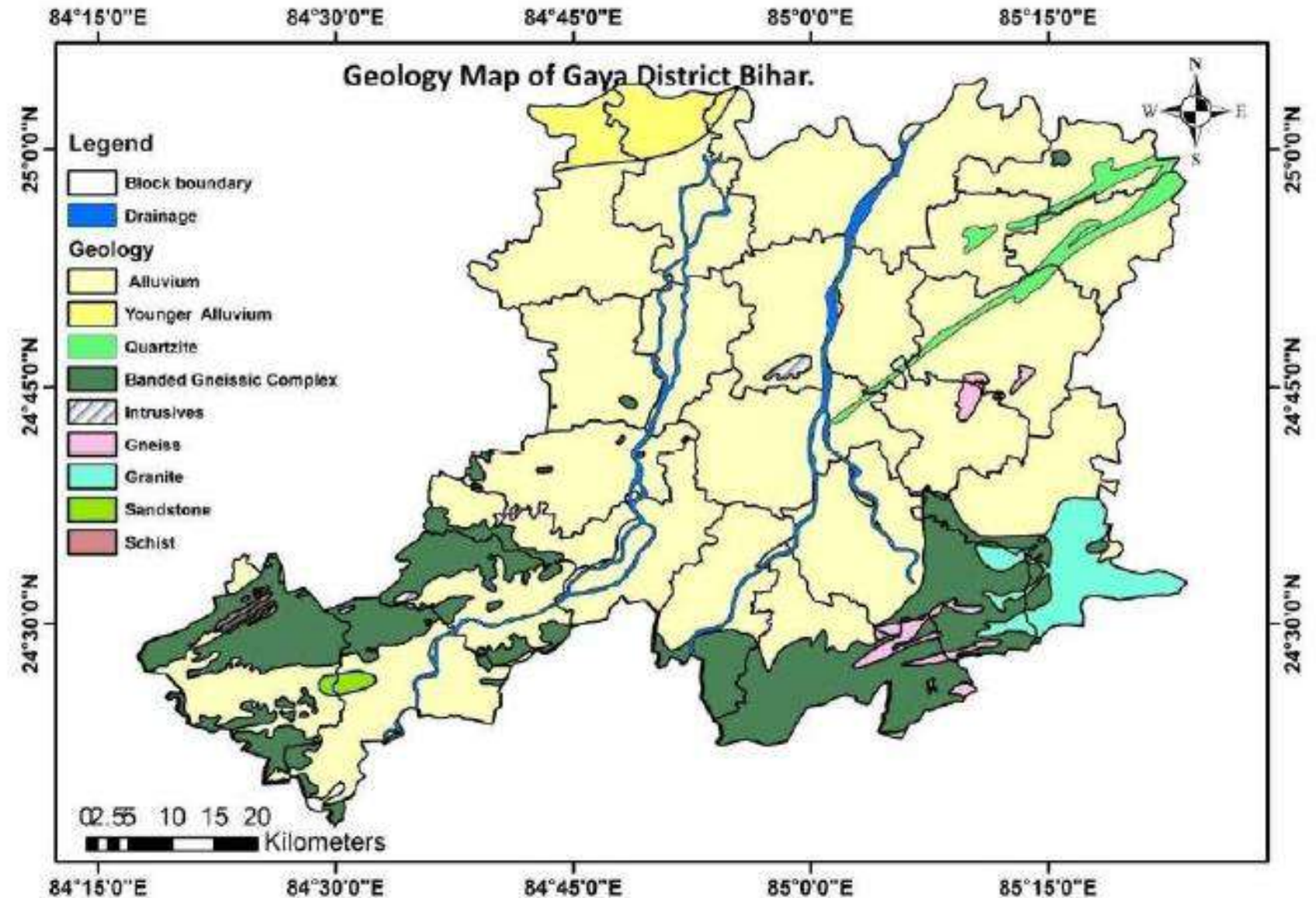


- Land use/Land Cover
- Catchment area occupied by agriculture land mainly



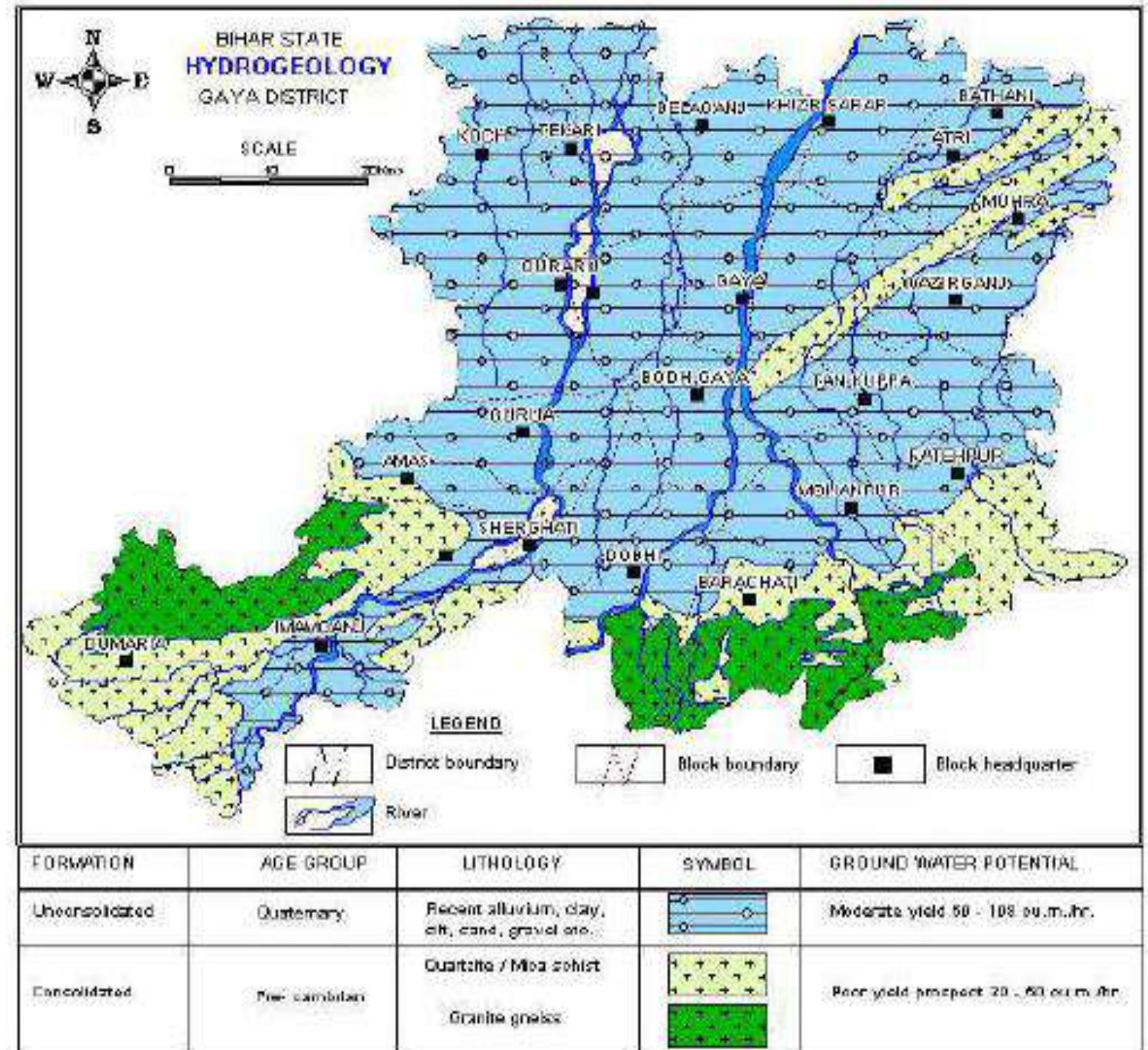
Sub-surface Geology

- Alluvium Deposit in the flood plain area
- In the origin area of river- Banded Gneissic Complex (BCG)
- BCG offers poor ground water storage potential, mainly secondary porosity



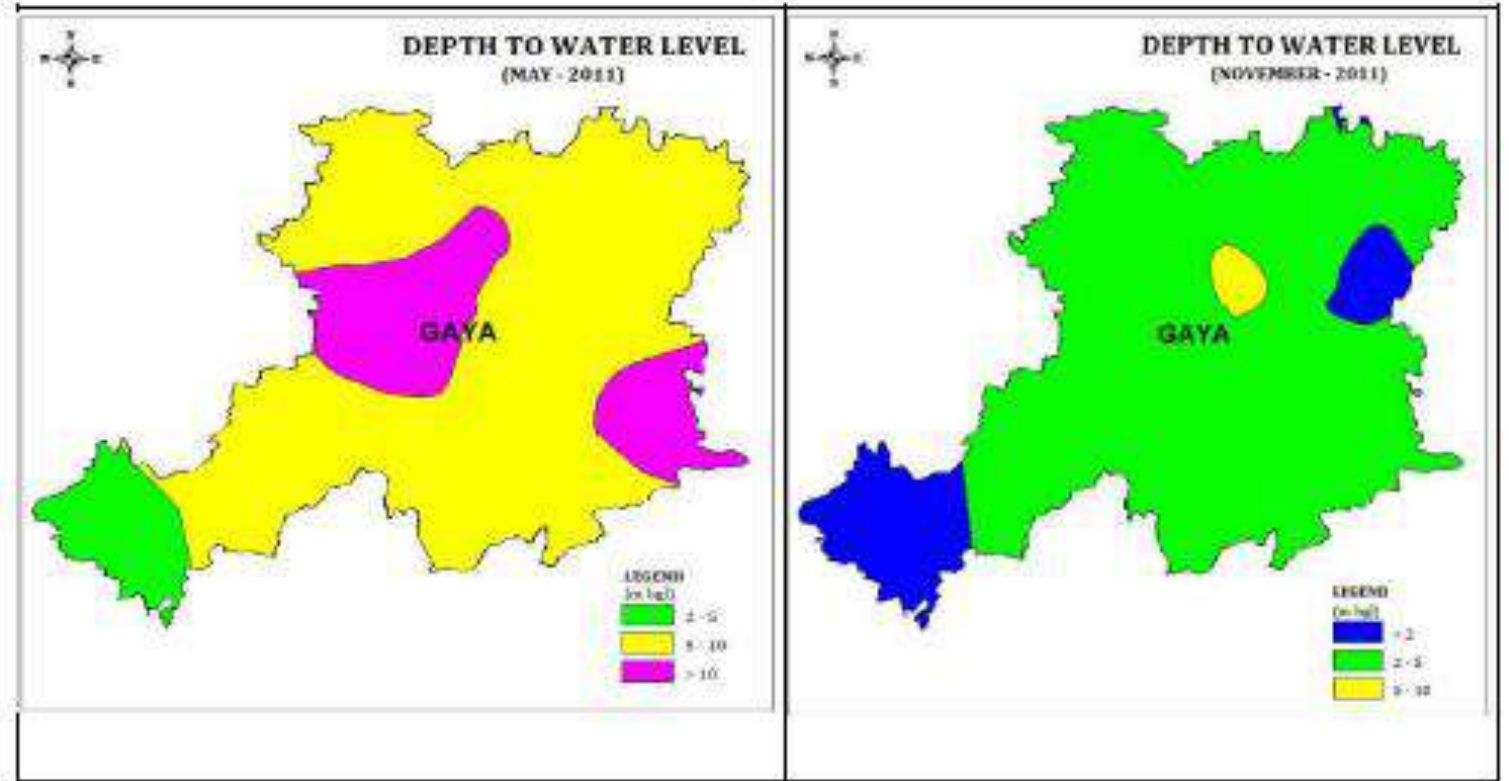
Hydrogeology

- Two hydrogeological unit.
- (A) Fissured formations, and
- (B) Porous formations
- Boreholes drilled by C.G.W.B. in hard rock terrain have tapped fractures at variable depth from 14 - 98 m. Yield varies 2.6 - 35.67 m³/hr (**POOR**).
- Transmissivity of fractures within 100 m depth in granite gneiss varies from 126-168 m²/day and storability varies from 1.2×10^{-3} - 1.4×10^{-5} . In Gaya district, **altogether 25 exploratory wells and 17 observation wells** have been drilled to decipher the potential fracture zones in hard rock formation.



Depth of Water Level

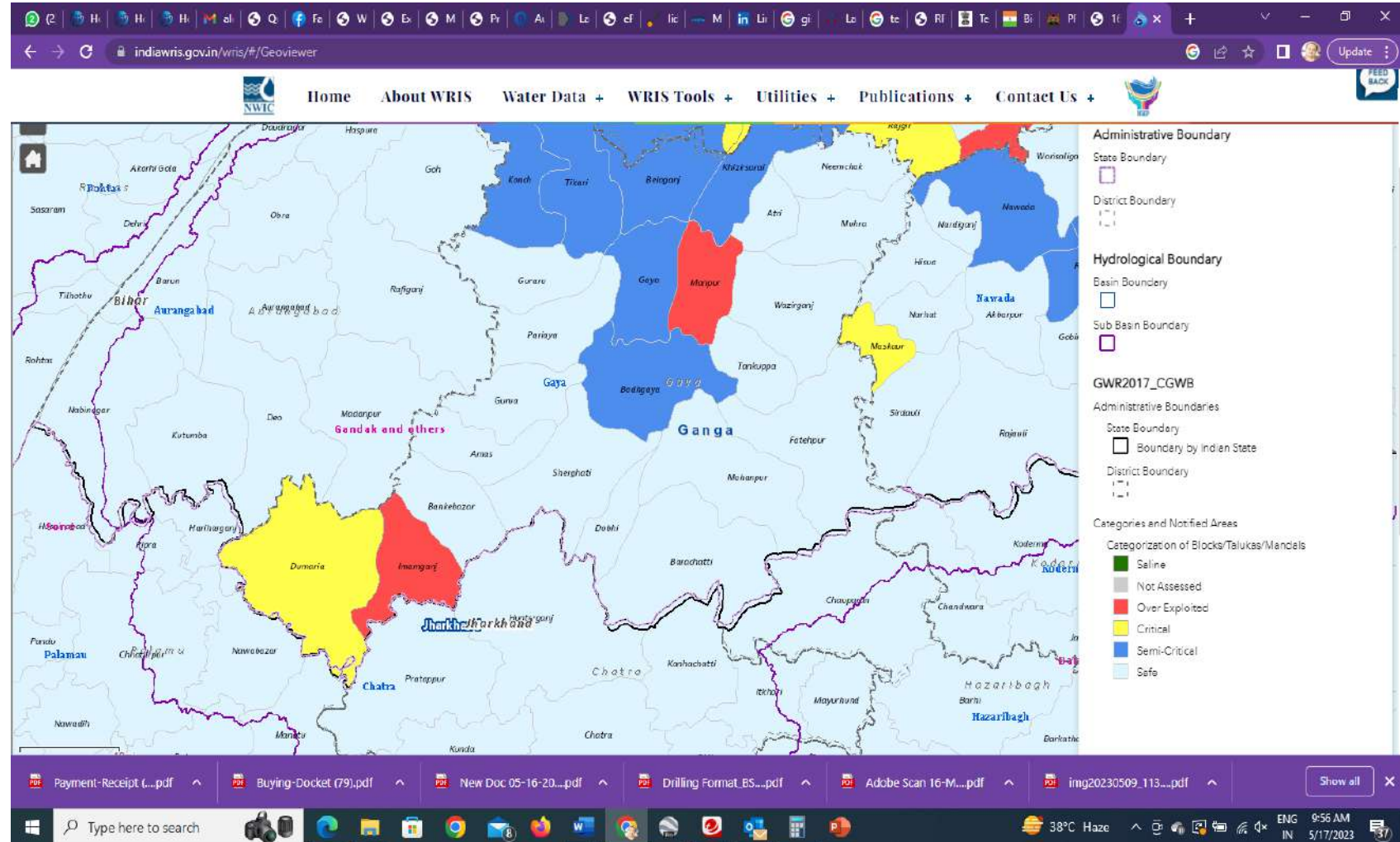
- Depth to water level in 2011 is 5-10 mtr. Data is not available on India WRIS
- There is post monsoon rise due to good storage availability



Depth of water table. (Pre monsoon and post Monsoon)

Ground water as per GWR 2017

- Gaya and Bodhgaya - **Semi Critical**
- Chatra Block - **Safe**



Key Issues

1. **Lack of proper data availability**, large scale Topographic, watershed, drainage etc.
2. **Lack of proper study like** Hydrogeological study, monitoring station, hydrological study, Gauge and discharge site, etc.
3. **Village and town effluent** are discharged to Niranjana (Phalgu) River.
4. **Farmers are withdrawing water** depleting the ground water table, need to do awareness for the conjunctive use of surface and ground water.
5. **Domestic Pollution:** River Niranjana (Phalgu) receives directly untreated domestic sewage from villages & cities.
6. **Encroachment of River Land:** as per the information provided by public during site visit river land has been encroached by local villagers and **used for farming by filling river land with sand or mud.**
7. **Reducing ground water levels:** Due to harnessing of ground water on large scale by domestic users, lack of rainfall and without proper management of water resources level of ground water is reducing.
8. **Lack of Planning, Design & Negligence:** Without proper catchment area, discharge calculations & without hydrology. Proper technology needs to be transferred for long term benefits of the river rejuvenation programme.
9. **Solid Waste Management:** **there is very limited solid waste management practices observed resulting in dumping of household waste and other solid waste on the banks of the nalla or river thereby choking the river flow.**
10. **Sand mining - major issue,** miners are digging illegally/ **with the tacit approval of government without any qualms and leaving it as dry pits.** Water layer is receding fast leaving the river sand less and waterless.
11. Origin of the Niranjana River is in hills, the elevation difference is quite high from origin to Vishnupad, so **slope of Niranjana River from origin to Vishnupad is steep**, due to this in the monsoon period water flows with high speed in whole river stretch, resulting into damage to both banks of the river.

Site Visits Photographs



Niranjana (Phalgu) River Near Gaya



Barrage of Lilajan Irrigation Scheme at Ghoraghat, Gaya



Main Canal from Barrage



Canal Exit at Head Regulator

Site Visits Photographs



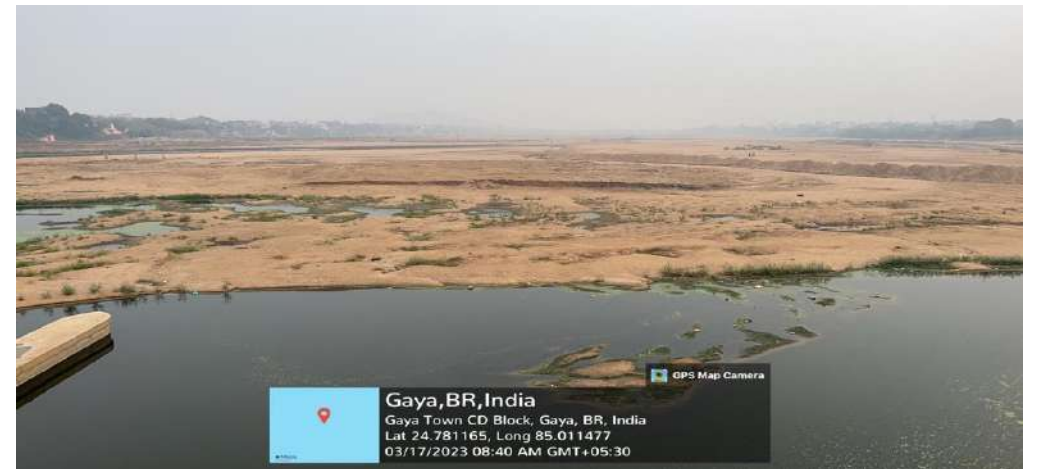
Niranjana (Phalgu) River at Vishnupad Ghaat



Crossing Bridge near Vishnupad Ghat on Niranjana (Phalgu) River



Current Construction Activities near Bridge



U/S view of River

NMCG Interventions

- **River boundary identification** from revenue records and maps.
- Identification of **low lying area** in river flood plain - Suitable **for construction of ponds for ground water recharge and base flow augmentation**. These activities can be taken up in new areas in phased manner.
- River channel is dry in most of the area or backfilled, further **cleaning and deepening of river channel** may be initiated with proper slope stabilization.
- Provision of **catchment area treatment** to increase inflow of clean water in river just like proper channelization, catch drains, silt traps etc.
- **Deepening and De-silting of existing ponds and river channel.**
- **Groundwater Recharge** (using recharge wells and bore wells)
- **Erosion control measure** (using boulders to deaccelerate the speed of river flow)
- Providing **solid waste management** at village level – avoid contamination and blockage.
- Change in cropping pattern & organic farming, encourage organic farming adjacent to river stream/area.

Way Forward

Water Availability Consideration

- **Survey and investigation, data collection, primary and secondary**
- Aerial LIDAR Survey of catchment area 3000 sq km for detailed topographic 3D mapping for River boundary, encroached land, low lying area in river flood plain
- **Primary data** collection such as inventory of borewells/ tube wells, wetlands, revenue records and maps, drains, drainage, point and non point source of pollution etc.
- Geophysical investigation – Electrical Resistivity Tomography
- Construction of exploratory bore wells and piezometer in catchment area for aquifer characteristics, pumping tests, hydrogeological studies Depth to water level, water table and fluctuation maps
- **Secondary data** collection- hydrometeorological data, other line departments
- Modelling- Ground water and surface water modelling for water availability and water balance
- DPR preparation of catchment area treatment of proposed cascading bunds, check dams, trenches etc.
- Ponds rejuvenation, large ponds for wetlands development-Fisheries and aquaculture/ livelihood promotion

Pollution Abatement

- **Natural waste water treatment for villages.** **STP not required** for city/town sewerage treatment
- Strict compliance for sewerage treatment before discharging into Niranjana (Phalgu) River.
- **Private Public Participation** - for sustainability of water availability and pollution abatement

Way Forward (Contd.)

- Report of the Niranjana river was discussed internally in NMCG.
- Course of action decided for the rejuvenation of the river is as follows:
 - Rejuvenation Plans shared with NRRM vide letter dated 22nd May, 2023
 - Interim immediate measures (possibly before onset of monsoon-2023) be taken-up with involvement of NRRM, HESCO and State Government. Concerned organizations to take up strategies to be adopted for rejuvenation of Phalgu river by taking up some interim measures in upper catchment.
 - Study, for long term impacts, be conducted to gather data.
 - NMCG may fund the study and findings to be implemented by State/ CGWB out of MNERGA/ Central/ State funds.

Tentative Scope of work for IIT Roorkee

- Data collection of primary and secondary source such as historical hydrometeorological data, historical ground water level/quality, ground water inventory, ground water draft, Evaporation from catchment, ERT, geological and geomorphological etc
- Delineation of watershed/watershed areas using aerial LIDAR photogrammetry, Topographic surveying and mapping of watershed/ watershed using manned, aerial LIDAR photogrammetry (GSD 10 cm) ground sampling distance is 10 cm
- Identification of Drainage pattern and estimation of surface water run off, inlets and outlets, infiltration studies etc in the watershed areas.
- Change detection /land use changes using high resolution images data
- Hydraulic modelling of surface water availability in the watershed area for the sustainable Falgu rejuvenation project.
- Drilling of Exploratory borewells in the watershed area/watershed for the aquifer characteristics. (15 numbers). Retrieved core will be utilized for weather history.
- Drilling and monitoring borewells (Piezometers) /observation wells and carrying out the pumping tests in the catchment/watershed. (15 number)
- Installation of digital water level recorders and setting up cloud based GW monitoring system
- Estimation of ground water reserves and potential/ water budget of watershed/catchment
- Site selection for Ground water augmentation through managed aquifer recharge and design of cascading artificial recharge structures, public utility, embankment etc
- Ground water flow modelling and development of scenarios for the water availability and water balance
- Species diversity and ecological surveys of catchment area, suggestions for further plantation to increase water holding and to check excessive evaporation
- Weather Station (Procurement) to be installed at site
- Bathymetric Survey, if required
- Recommendations and strategies (All studies, analysis, Economic Impact Assessment etc.,
- DPR Preparation, bill of quantities and tender doc preparation and construction supervision

Immediate Measures that may be taken are:

IS 15792: 2008 Artificial Recharge to Ground Water

- Bench Terracing
- Contour Bunds
- Contour Trenches
- Gully Plugs, Nala Bunds & Check Dams
- Percolation Tanks/ Ponds

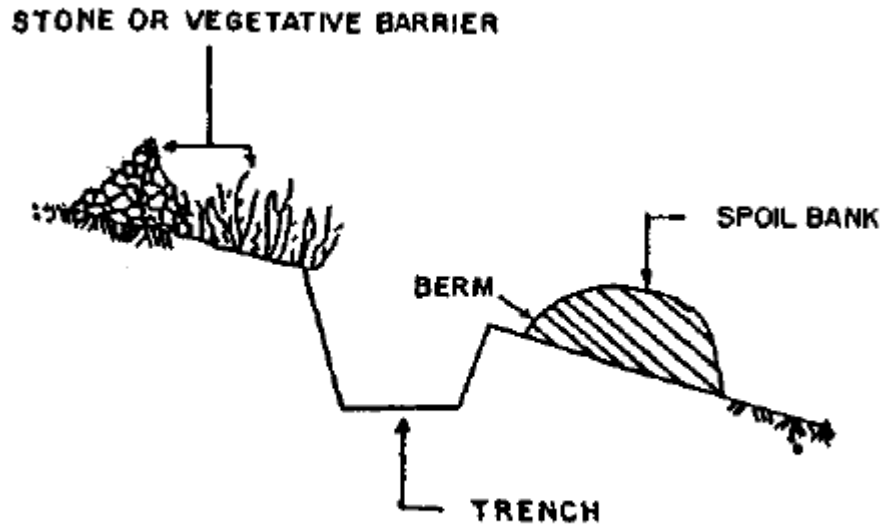
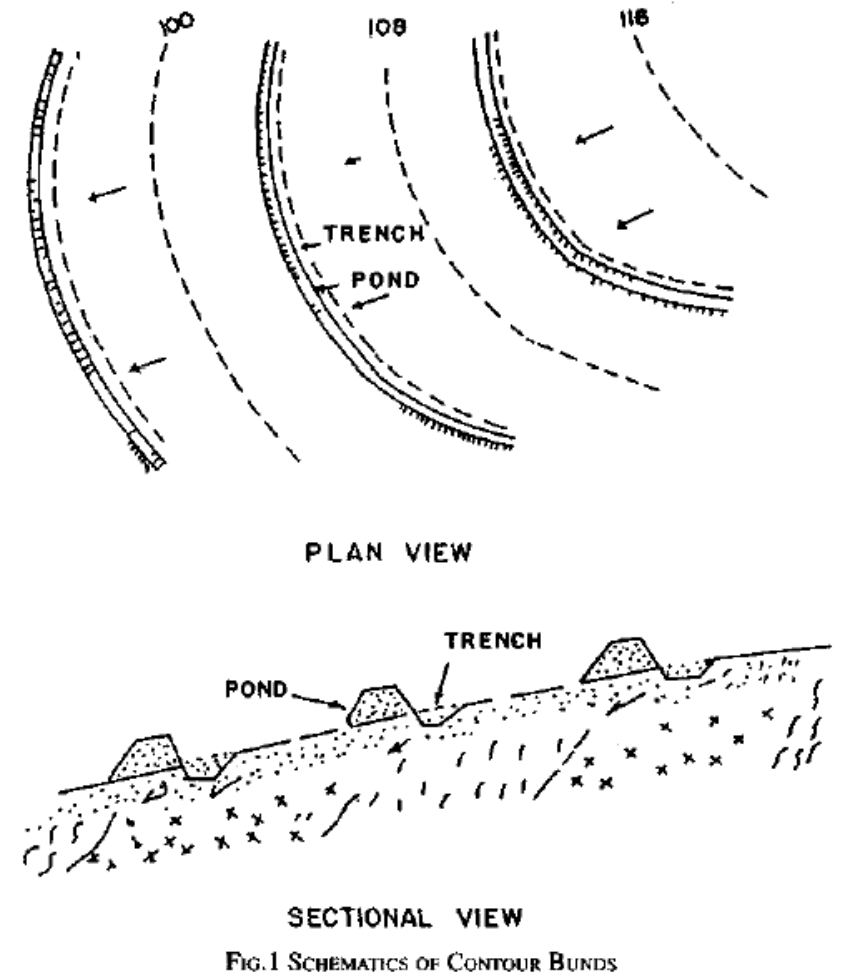


FIG. 2 SCHEMATICS OF A CONTOUR TRENCH



Artificial Recharge Techniques

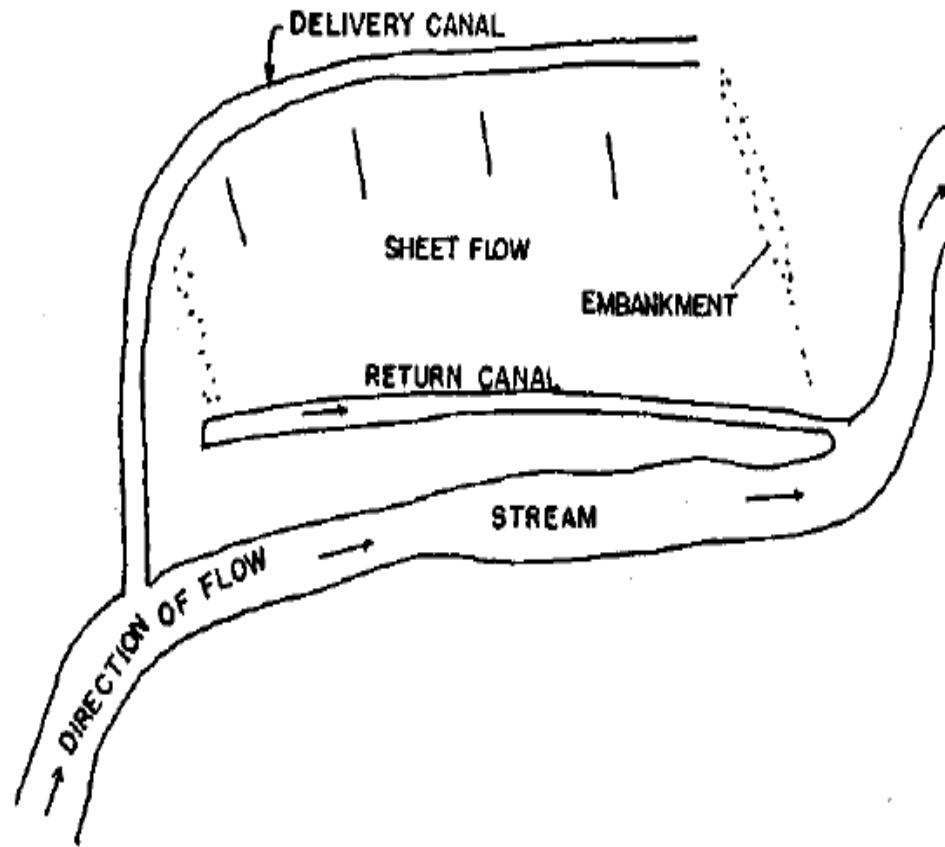


FIG. 5 SCHEMATICS OF A TYPICAL FLOOD RECHARGE SYSTEM

4.1 Direct Methods

4.1.1 Surface Spreading Techniques

a) Runoff Conservation Structure:

- 1) Bench terracing
- 2) Contour Bunds
- 3) Gully Plugs, Nala Bunds, Check Dams
- 4) Percolation Ponds

b) Flooding;

c) Ditch and Furrows;

d) Recharge Basins; and

e) Stream Modification/Augmentation.

4.1.2 Sub-surface Techniques

a) Injection wells (Recharge wells),

b) Gravity head recharge wells (Dug well/Bore well/Tube well recharge), and

c) Recharge pits and shafts.

4.1.3 Combination Techniques

These include combination of surface and sub-surface techniques.

4.2 Indirect Methods

a) Induced recharge from surface water sources, and

b) Aquifer modification:

- 1) Bore Blasting, and
- 2) Hydro-fracturing.

Artificial Recharge Methods During Flood Time

IS 15792 : 2008

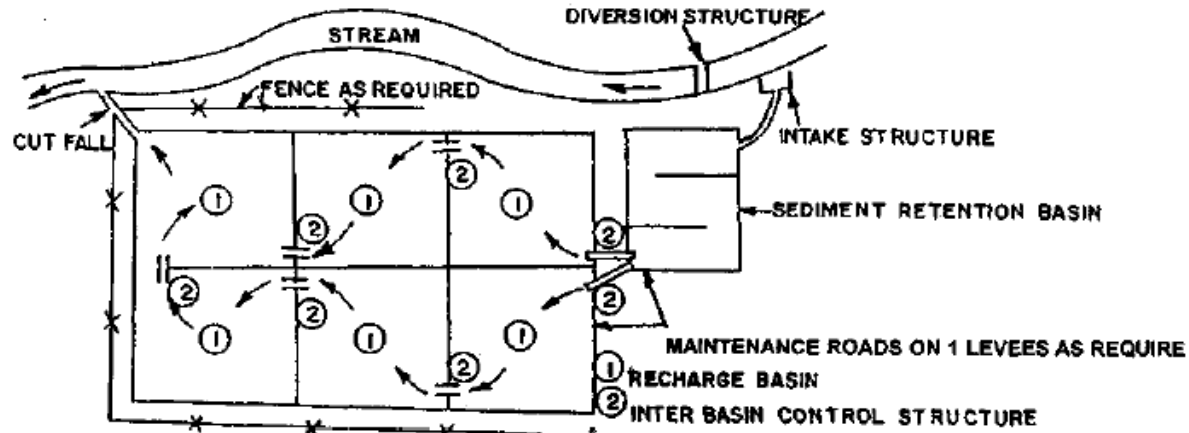


FIG. 7 SCHEMATICS OF A TYPICAL RECHARGE BASIN

- Injection Wells or Recharge Wells
- Recharge Pits
- Recharge Shafts
- Sub Surface Dykes/ Ground Water
- Dams/ Underground 'Bhandaras'

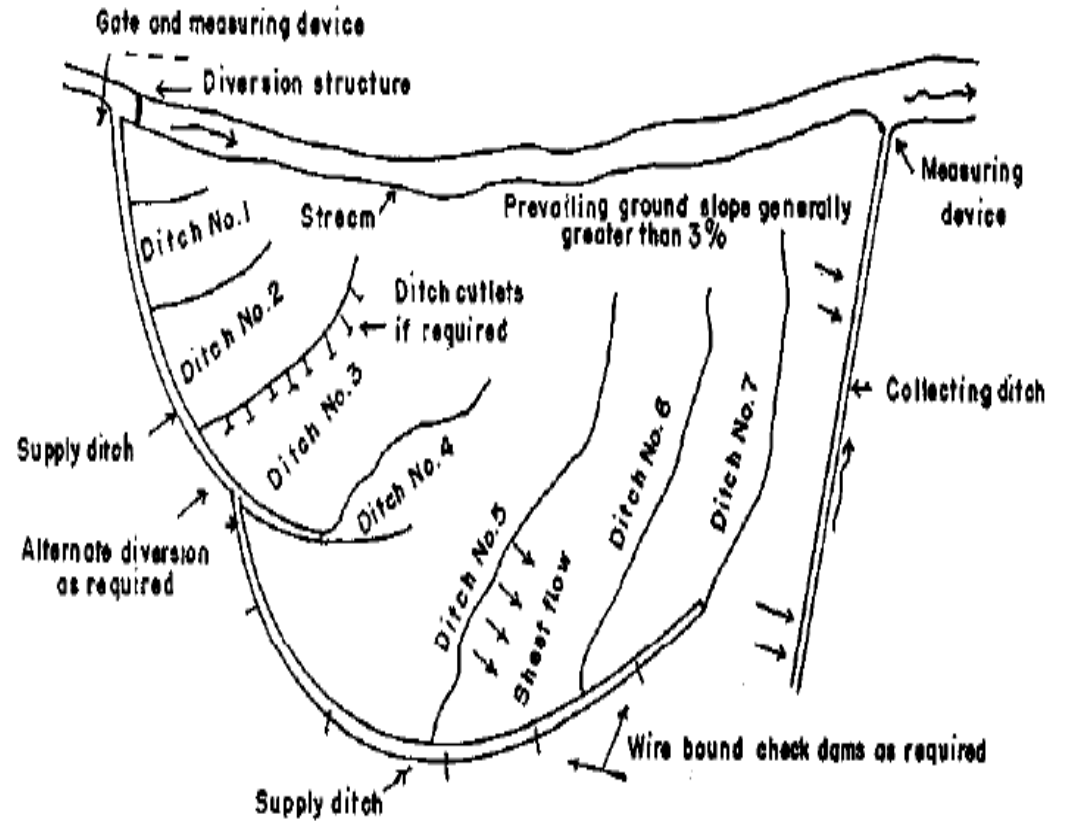


FIG. 6 DITCH AND FURROW METHOD

Trench Details		
Trench No.	Shape	Length (m)
1	Polyline	25
2	Polyline	26
3	Polyline	61
4	Polyline	154
5	Polyline	174
6	Polyline	116
7	Polyline	220
8	Polyline	107
9	Polyline	360
10	Polyline	312
11	Polyline	289
12	Polyline	656
13	Polyline	586
14	Polyline	364
15	Polyline	410
16	Polyline	421
17	Polyline	372
18	Polyline	383
19	Polyline	297
20	Polyline	239
21	Polyline	313
22	Polyline	254
23	Polyline	204

24	Polyline	296
25	Polyline	650
26	Polyline	600
27	Polyline	568
28	Polyline	421
30	Polyline	228
31	Polyline	240
32	Polyline	294
33	Polyline	166
34	Polyline	267
35	Polyline	255
36	Polyline	381
37	Polyline	602
38	Polyline	206
39	Polyline	153
40	Polyline	157
41	Polyline	144
41	Polyline	130
42	Polyline	176
42	Polyline	112
43	Polyline	160
43	Polyline	295
44	Polyline	148
44	Polyline	178
44	Polyline	169
45	Polyline	376
46	Polyline	168

47	Polyline	403
48	Polyline	402
49	Polyline	311
50	Polyline	175
51	Polyline	109
52	Polyline	209
53	Polyline	515
54	Polyline	218
55	Polyline	224
56	Polyline	252
57	Polyline	258
58	Polyline	384
59	Polyline	274
60	Polyline	223
61	Polyline	295
62	Polyline	298
63	Polyline	244
64	Polyline	111
65	Polyline	95
66	Polyline	87
67	Polyline	86
68	Polyline	67
69	Polyline	69
70	Polyline	141
71	Polyline	89
72	Polyline	110
73	Polyline	137
74	Polyline	89
75	Polyline	207
76	Polyline	163
77	Polyline	119
78	Polyline	120
79	Polyline	110
80	Polyline	136
81	Polyline	144

82	Polyline	297
83	Polyline	179
84	Polyline	165
85	Polyline	181
86	Polyline	203
87	Polyline	175
88	Polyline	209
89	Polyline	85
90	Polyline	160
91	Polyline	136
91	Polyline	229
92	Polyline	273
93	Polyline	200
93	Polyline	163
94	Polyline	161
95	Polyline	182
96	Polyline	261
96	Polyline	243
97	Polyline	171
98	Polyline	144
99	Polyline	115
100	Polyline	134
101	Polyline	120
102	Polyline	141
103	Polyline	168
104	Polyline	141
105	Polyline	145
106	Polyline	197
107	Polyline	226
108	Polyline	114
109	Polyline	124
110	Polyline	143
110	Polyline	218
111	Polyline	105
112	Polyline	195
113	Polyline	144
114	Polyline	163
115	Polyline	155
116	Polyline	117
Total		27438
		27 Kms

THANKS